

Lecture 1: Linear Models & ML Notation

1. The big picture

Machine Learning is often about using **input data** to **predict an output**.

For example:

```
Wind speed  -----+
                                   +----> Model ----> Power output
Temperature  -----+
```

A linear model represents this relationship mathematically:

$$\hat{y} = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$$

The important idea is not just the equation. It is:

We have data, we choose a mathematical model, and the model learns useful parameters from the data so it can make predictions.

2. Why do we need ML if formulas already exist?

If we already know the exact formula for a problem, we usually **do not need ML**.

For example:

$$\text{Area} = \pi r^2$$

If we know the radius, we can calculate the exact area directly.

ML becomes useful when:

- the relationship is **unknown**
- the relationship is too **complicated** to describe with a simple formula
- we have lots of examples/data from which the relationship can be learned

For example, predicting house prices:

```
House size
Number of rooms
Location
Age
Distance from metro
...
      v
    ??? relationship ???
      v
    House price
```

Instead of manually creating an exact formula, we can give the model many examples and let it learn parameters such as the weights.

Key idea

Known formula -> calculate the answer.

Unknown/complex relationship + data -> ML can learn an approximation.

3. Feature vs Example

This distinction is fundamental.

Imagine a dataset:

Example	Size	Bedrooms	Age	Price
1	1200	2	10	Rs 60L
2	1800	3	5	Rs 95L
3	900	2	20	Rs 45L
4	1500	3	8	Rs 80L

Feature

A **feature** is one type of input information.

Examples:

- Size
- Bedrooms
- Age

Think:

Feature = column / type of input information

Example

An **example** is one complete observation/data point.

Example 1 contains:

1200 sq ft
2 bedrooms
10 years old
Rs 60L

Think:

Example = row / one observation

So:

Feature → WHAT kind of information?
Example → WHICH data point?

4. The most important notation: $x_j^{(i)}$

The notation

$$x_j^{(i)}$$

means:

feature j of example i

The two indexes have different jobs:

x
|
subscript j
v
WHICH FEATURE?

superscript (i)
v
WHICH EXAMPLE?

Example

$$x_2^{(3)}$$

Read it as:

"x sub 2, superscript 3"

But when thinking about its meaning, translate it to:

feature 2 of example 3

Using the table above:

- feature 2 = Bedrooms
- example 3 = the third row
- therefore:

$$x_2^{(3)} = 2$$

Important

The superscript (i) here is **NOT** an exponent.

$$x_j^{(i)}$$

is using the superscript as an index/label for the example number.

5. What is $y^{(i)}$?

The symbol

$$y^{(i)}$$

means:

the actual target/output of example i

Read it as:

"y of example i"

There is no feature number because y represents the target itself.

For example:

$$y^{(3)} = \text{Rs } 45\text{L}$$

means:

the actual price (target) of example 3 is Rs 45L.

So:

$$x_2^{(3)}$$

-> feature 2 of example 3

$$y^{(3)}$$

-> actual output of example 3

6. What is $\hat{y}$?

The symbol

$$\hat{y}$$

is read:

"y-hat"

It represents the **model's prediction**.

The difference is:

y = actual value

$\hat{y}$ = predicted value

For example:

Actual house price

y
 $y = \text{Rs } 60\text{L}$

Model prediction

$\hat{y}$
 $\hat{y} = \text{Rs } 57\text{L}$

So:

$\hat{y}^{(i)}$

means:

the model's predicted output for example i

Read it as:

"y-hat of example i."

7. Linear model with multiple features

For one input feature:

$$\hat{y} = wx + b$$

For multiple input features:

$$\hat{y} = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$$

Each feature is multiplied by its corresponding weight, then everything is added together with the bias.

For example:

$$\hat{y} = 100x_1 + 20x_2 + 50$$

Here:

- $w_1 = 100$
- $w_2 = 20$
- $b = 50$

The model uses these values to produce a prediction.

8. What are weights?

A **weight** tells the model how strongly a feature contributes to the prediction.

In:

$$\hat{y} = w_1x_1 + w_2x_2 + b$$

- w_1 belongs to x_1
- w_2 belongs to x_2

In machine learning, these weights are typically **learned from training data** rather than manually chosen.

9. What is b ?

b is called the **bias**.

For now, the easiest way to think about it is as the **constant/intercept** in a line.

Compare:

$$y = mx + c$$

from school with:

$$\hat{y} = wx + b$$

So initially:

b = bias / intercept

There is no need to go deeper into bias yet.

10. What do m and n mean?

These are easy to confuse.

m = number of training examples

If our dataset has 1,000 rows:

$$m = 1000$$

n = number of input features

If every example has 5 input features:

$$n = 5$$

Remember:

m -> how many examples/rows?

n -> how many features/inputs?

11. Putting the notation together

For training example i , the linear model is:

$$\hat{y}^{(i)} = w_1 x_1^{(i)} + w_2 x_2^{(i)} + \dots + w_n x_n^{(i)} + b$$

Read the equation in plain English:

For example i , take each input feature, multiply it by its corresponding weight, add them together, add the bias, and get the model's predicted output.

The actual output is:

$$y^{(i)}$$

The prediction is:

$$\hat{y}^{(i)}$$

Eventually, ML will compare these two:

$$y^{(i)} \quad \text{vs} \quad \hat{y}^{(i)}$$

to determine how good the prediction is.

12. A complete example

Suppose:

$$x_1 = \text{wind speed}$$

$$x_2 = \text{temperature}$$

and:

$$\hat{y} = w_1 x_1 + w_2 x_2 + b$$

For example 1:

$$x_1^{(1)} = 5$$

$$x_2^{(1)} = 20$$

and the actual power output is:

$$y^{(1)} = 1500$$

The model produces:

$$\hat{y}^{(1)} = w_1(5) + w_2(20) + b$$

For example 2:

$$x_1^{(2)} = 7$$

$$x_2^{(2)} = 25$$

and:

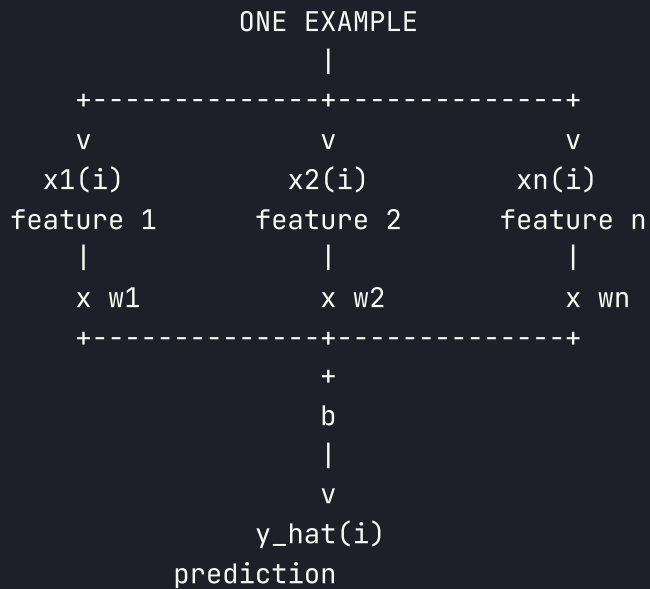
$$\hat{y}^{(2)} = w_1(7) + w_2(25) + b$$

Notice how the **same weights and bias** are used, but the input values change from one example to another.

Quick Revision Sheet

Symbol	Meaning	How to read/think
x_j	j-th input feature	feature j
$x_j^{(i)}$	feature j of example i	"feature j of example i"
$y^{(i)}$	actual target of example i	"y of example i"
$\hat{y}$	model prediction	"y-hat"
$\hat{y}^{(i)}$	prediction for example i	"y-hat of example i"
w_j	weight for feature j	importance/contribution parameter
b	bias	intercept/constant
m	number of examples	number of rows
n	number of features	number of input columns

The one mental model to remember



Actual target:
y(i)

Remember these three translations

$$x_j^{(i)} = \text{feature } j \text{ of example } i$$

$$y^{(i)} = \text{actual output of example } i$$

$$\hat{y}^{(i)} = \text{predicted output of example } i$$

And the central idea of the lecture:

$$\text{Data} \rightarrow \text{Model} \rightarrow \text{Prediction}$$

ML is useful when we want a model to **learn a relationship from data**, especially when we don't already have a simple exact formula for that relationship.